Due: August 12, 2026

1. Solve the following equations:
   a) $4x + 7 = 12$
   b) $2x + 6 = 3x - 5$
   c) $14x - 10 = 6x - 8$
2. Solve the following inequalities:
   a) $2x + 5 < x - 4$
   b) $2x - 1 < 6x + 5$
   c) $|4 - 3x| < 2$
3. If $7x - 3 < 0$ and $7x > 0$, express these as a compound inequality and find its solution.
4. Expand the following summation expressions:
   a)
   $$\sum_{i=5}^{8} a_i x_i$$
   b)
   $$\sum_{i=1}^{n} a_i x^{i-1}$$
5. Rewrite the following in summation notation:
   a) $x_1(x_1 - 1) + 2x_2(x_2 - 1) + 3x_3(x_3 - 1)$
   b) $\frac{1}{x} + \frac{1}{x^2} + ... + \frac{1}{x^n}$ $(x \neq 0)$
6. Solve the following polynomial by factoring:
   a) $x^2 + x - 6 = 0$
   b) $x^3 + 2x^2 - 4x - 8 = 0$
   c) $x^2 - 9 = 0$
7. Find the zeros of the following functions by the quadratic formula (show work)
   a) $f(x) = x^2 - 7x + 10$
   b) $g(x) = 2x^2 - 4x - 16$
8. Find a cubic function with roots 4, -2, 3.
9. What are the values of the following logarithms?

a) $\log_{10} 0.0001$

b) $\log_5 3125$

10. Evaluate the following:

a) $\ln e^7$

b) $\ln \frac{1}{e^5}$

c) $\log_e e^{-4}$

11. Evaluate the following by application of the rules of logarithms:

a) $\ln Ae^2$

b) $\ln ABe^{-4}$

12. Evaluate the following expressions:

a) $4!$

b) ${}_6P_2$

c) ${}_6C_2$

13. Suppose the probability that a country is a democracy is $P(D) = 0.4$, the probability that a country experiences a civil war is $P(W) = 0.15$, and the probability of both occurring is $P(D \cap W) = 0.05$.

a) What is the probability that a country is NOT a democracy?

b) What is the probability that a country is either a democracy OR experiences a civil war (or both)?

14. A political analyst uses a warning system to predict coups. The probability of a coup occurring in any given country-year is $P(C) = 0.05$. If a coup occurs, the warning system triggers a warning with probability $P(W \mid C) = 0.80$. If no coup occurs, the system triggers a warning with probability $P(W \mid C^c) = 0.10$.

a) What is the total probability that the system triggers a warning in a given year? (Hint: Use the Law of Total Probability).

b) If the warning system triggers a warning, what is the probability that a coup actually occurs? (Hint: Use Bayes' Rule).